# CHAPTER 8

# MAGNETIC FIELD

## Q. 1. What is magnetism? Write its various properties and uses?

### MAGNETISM

The phenomenon of magnetism has been known of since ancient times. The Greeks, Romans and Chinese knew the mineral lodestone, an oxide of iron that has the property of attracting iron objects.

The body which attracts small pieces of iron and points north south when suspended freely is called magnet and its power of attraction is called magnetization. The study of properties associated with magnets is called magnetism.

### TYPES OF MAGNETS

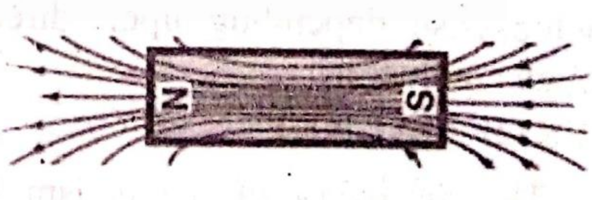
The magnets have two types called permanent magnets and electromagnets.

### 1: PERMANENT MAGNET

The permanent magnetic material comes from ores. The permanent magnets are also made from hard magnetic materials such as cobalt, steel or alnico (alloy), which are magnetized by electromagnetic induction. Following are the main properties of permanent magnets.

### PROPERTIES OF PERMANENT MAGNETS

1. When a bar magnet is dipped into iron fillings, few of them adhere to central part while many of them cluster to ends. The ends are called poles of the bar magnet.
2. The magnet has two poles called north pole and south pole.
   The north pole and south pole attracts each other.

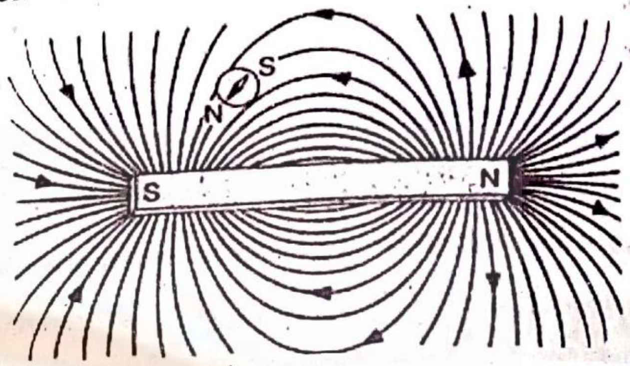



The north pole repels another north pole while a south pole repels another south pole.

❸ The pole of a magnet never be separated from the magnet. The separated end is again magnet when it is separated. The magnetization is concentrated at poles than central part of the magnet.

❹ The steel or iron bar gets magnetized when it is rubbed against the magnet. Heating or dropping it on ground several times can demagnetize it.

❺ The space or region around a magnet in which attraction or repulsion with other magnets takes place is called magnetic field. It is denoted by $\vec{B}$. The magnetic field is a vector quantity. Its direction is defined by magnetic lines of force.




The magnetic lines of force come out from north pole

The magnetic lines of force enter into south pole.

## 2: ELECTROMAGNET

The magnetic field is generated around a straight conductor in the form of circles when current flows through it. This conductor is called electromagnet.

## ELECTROMAGNETIZATION

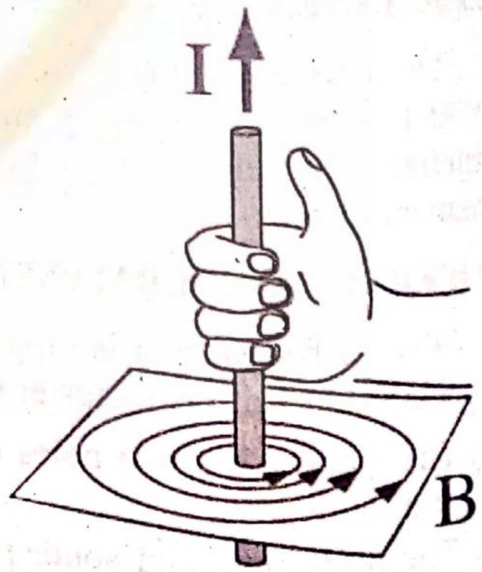
The power of attraction of electromagnet is called electromagnetization.

## ELECTROMAGNETISM

The study of properties associated with electromagnet is called electromagnetism.

## ELECTROMAGNETIC FIELD

The magnitude of generated electromagnetic field depends upon the amount of current in conductor whose value is given by Ampere law. The direction of electromagnetic field is given by right hand rule.




## RIGHT HAND RULE

Hold the current carrying conductor in your right hand in such a way that erect thumb is along the direction of current. The curled fingertips indicate the direction of magnetic field which will be clockwise or anticlockwise depending upon direction of current through conductor.

## USES OF MAGNET

❶ The discovery of magnetism has a great practical use from small refrigerator magnets to magnetic recording tape and computer disks.

❷ The magnetism of individual atomic nuclei is used by physicians to make images of organs deep within the body.

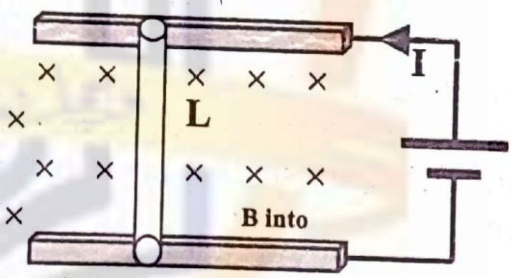
- Spacecrafts have measured the magnetism of the earth and the other planets to learn about their internal structure.
- The electromagnet is used in electric bells, electric motors, fans, radio, telephone, galvanometer, transformer etc.
- Heavy pieces of iron can be lifted and shifted from one place to an other place by electromagnet.
- Doctors use electromagnet to remove iron fillings from eyes and bullets from wounds.
- The electromagnet is used in cyclotron accelerator.

**Q.2. Derive a formula for magnetic force acting on current carrying conductor placed in a magnetic field.**

## MAGNETIC FORCE ON CURRENT CARRYING CONDUCTOR PLACED IN A MAGNETIC FIELD

Consider a conductor having length L is placed over a conducting rail in applied magnetic field B perpendicularly. The magnetic field is generated around the conductor in the form of circles when current I flows through conductor. The generated magnetic field and applied magnetic field interacts together and give rise to magnetic force.

It is experimentally observed that motion of conductor on rail becomes fast when amount of current through circuit or applied magnetic field increases. It is also observed that motion of conductor increases when length of conductor increases. Similarly, motion of conductor is also affected with its orientation to magnetic filed.




This magnetic force $F_m$ acting on conductor is directly proportional to current I and length L. The magnetic force is also proportional to applied magnetic field B and orientation (Sinθ) of conductor with magnetic field.

$$F_m \propto I$$
$$F_m \propto L$$
$$F_m \propto B$$
$$F_m \propto \sin\theta$$

On combining al relations

$$F_m \propto I\,L\,B\,\sin\theta$$
$$F_m = k\,I\,L\,B\,\sin\theta$$

Where k is constant of proportionality. Its value in SI system is unity.

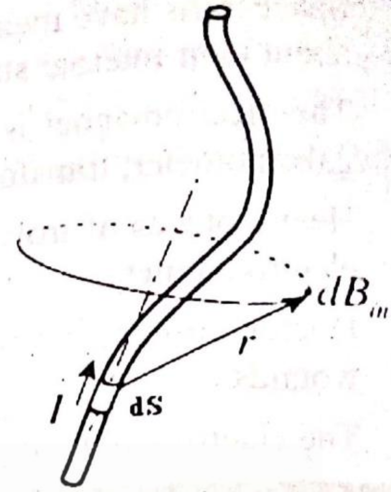
$$F_m = I\,L\,B\,\sin\theta$$

In vector form, the magnetic force is

$$\vec{F}_m = I(\vec{L} \times \vec{B}) = I\,L\,B\,\sin\theta\,\hat{n}$$

Where $\vec{L}$ is called vector length of conductor whose magnitude is the length of the conductor and direction is the direction of current.

When wire is not straight or magnetic field is non-uniform, then to calculate magnetic force, divide the wire into small segments of length dS for which length becomes uniform approximately.




The magnetic force on each segment is

$$d\vec{F}_m = I(d\vec{S} \times \vec{B})$$

The net magnetic force on conductor having length L is

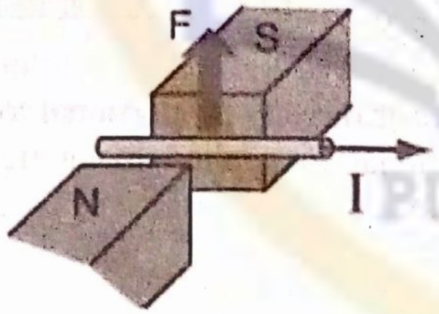
$$\vec{F}_m = \int_0^L I(d\vec{S} \times \vec{B})$$

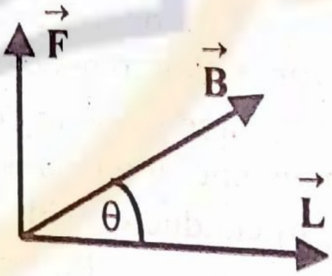
$$\vec{F}_m = \int_0^L I(d\vec{S} \times \vec{B})$$

## DIRECTION OF MAGNETIC FORCE

The direction of magnetic force is given by right hand rule stated as

Place close-fisted right hand at a location where tails of length vector $\vec{L}$ and magnetic field vector are joined together in such a way that fingertips are along the direction of rotation through smaller angle, then stretched thumb indicates the direction of magnetic force.




## Q.3 Calculate magnetic force on a moving charge in a magnetic field.

## MAGNETIC FORCE ON A MOVING CHARGE IN A MAGNETIC FIELD

Consider a conductor having length L and cross sectional area A and place it in a uniform magnetic field B perpendicularly as shown in fig. The volume of a conductor is AL. The conductor has many charge carriers.

## CHARGE CARRIERS

The conductor has many charge carries.

Unit volume of conductor has charge carriers = n

AL volume of conductor has charge carriers = nAL

Total charge carriers = nAL

## TOTAL CHARGE

One charge carrier has charge $= q$

All charge carriers have charge $= nAL$

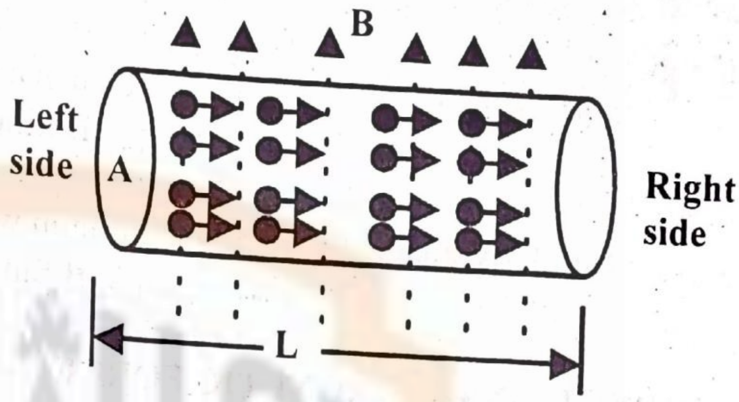
Total charge on all charge carriers is

$$\Delta Q = n\,A\,L\,q$$

## CURRENT FLOWING THROUGH CONDUCTOR

The charge carriers move from left side of a conductor with velocity V and come out at right side after covering distance L in time Δt given as

$$\Delta t = \frac{L}{V}$$




The current established within conductor is given as

$$I = \frac{\Delta Q}{\Delta t}$$

$$I = \frac{n\,A\,L\,q}{L/V}$$

$$I = n\,A\,q\,V$$

The magnetic force on current carrying conductor having length L is

## MAGNETIC FORCE ON CURRENT CARRYING CONDUCTOR

The magnetic force $\vec{F}_L$ on current carrying conductor having length vector $\vec{L}$ placed in applied magnetic field $\vec{B}$ is

$$\vec{F}_L = I(\vec{L} \times \vec{B})$$

$$\vec{F}_L = n\,A\,qV(\vec{L} \times \vec{B})$$

$$\vec{F}_L = n\,A\,qV(L\hat{L} \times \vec{B})$$

Where $\hat{L} = \hat{V}$ because direction of length vector and velocity vector is same.

$$\vec{F}_L = n\,A\,qV(L\hat{V} \times \vec{B})$$

$$\vec{F}_L = n\,A\,q\,L(V\hat{V} \times \vec{B})$$

$$\vec{F}_L = n\,A\,q\,L(\vec{V} \times \vec{B})$$

## MAGNETIC FORCE ON SINGLE CHARGE

The magnetic force $\vec{F}_m$ on a single charge q moving with velocity $\vec{V}$ in applied magnetic field $\vec{B}$

$$\vec{F}_m = \frac{\vec{F}_L}{n\,A\,q\,L} = \frac{n\,A\,q\,L(\vec{V} \times \vec{B})}{n\,A\,q\,L}$$

$$\vec{F}_m = q(\vec{V} \times \vec{B})$$

This is magnetic force on single charge which enters in applied magnetic field. It can be written as

$$\vec{F}_m = q\,V\,B\,\mathrm{Sin}\theta\ \hat{n}$$

## MAGNITUDE OF MAGNETIC FORCE

The magnitude of magnetic force is

$$F_m = q\,V\,B\,\mathrm{Sin}\theta$$

The magnetic force is maximum when angle between V and B is $90^0$.

$$F_{max} = q\,V\,B\,\mathrm{Sin}90^0$$
$$F_{max} = q\,V\,B$$

The magnetic force is minimum when angle between V and B is $0^0$

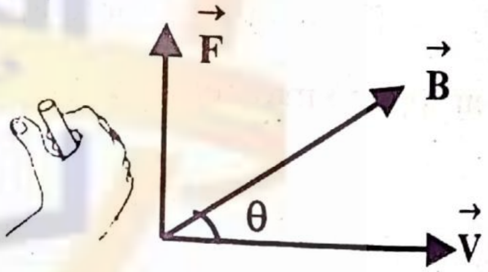
$$F_{min} = q\,V\,B\,\mathrm{Sin}0^0$$
$$F_{min} = 0$$

Similarly, magnetic force is minimum when angle between V and B is $180^0$

$$F_{min} = q\,V\,B\,\mathrm{Sin}180^0$$
$$F_{min} = 0$$

## DIRECTION OF MAGNETIC FORCE

The direction of magnetic force $\vec{F}_m$ is given by right hand rule stated as "join the tails of velocity vector $\vec{V}$ of +q charge and magnetic field vector $\vec{B}$ together. This defines a plane. Place close-fisted right hand at a location where tails of two vectors be joined in such a way that finger tips are along rotation through smaller angle. The erect thumb indicates the direction of magnetic force




$$\vec{F}_m = q(\vec{V}\times\vec{B})$$

The direction of magnetic force is reversed when charge is negative.

## SECOND DEFINITION OF RIGHT HAND RULE

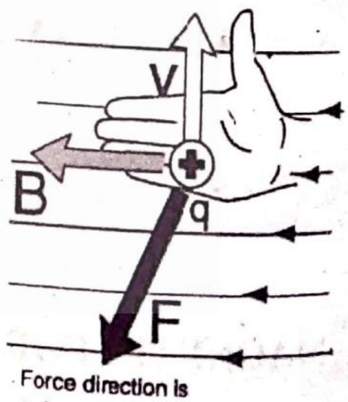
The right hand rule can also be stated as

❶ Point the thumb of your right hand in the direction of velocity $\vec{V}$ of +q charge

❷ Fingers in the direction of magnetic field $\vec{B}$.

❸ Palm direction is the direction of magnetic force

$$\vec{F}_m = q(\vec{V}\times\vec{B}).$$




Force direction is outward from palm.

## MAGNETIC FORCE ON ELECTRON

The magnetic force acting on an electron entering with velocity V in applied magnetic field B is

$$\vec{F}_m = -e(\vec{V}\times\vec{B})$$

## MAGNETIC FORCE ON PROTON

The magnetic force acting on proton entering with velocity v in applied magnetic field B is

$$\vec{F}_m = +e(\vec{V} \times \vec{B})$$

When an electron and proton are projected into magnetic field in the same direction both are deflected in opposite direction because magnetic force acts on them in opposite direction.

## Q.4 Discuss motion of charged particle in an electric field and magnetic field, hence define Lorenz force?

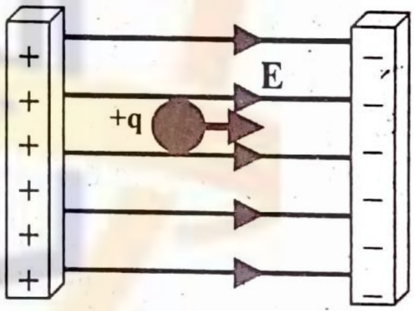
### MOTION OF CHARGED PARTICLE IN AN ELECTRIC FIELD

Consider a + q charge is placed in a uniform electric field. The electric force acting on the charge is given as

$$\vec{F}_e = q\vec{E} \quad \text{-------- (1)}$$

The charge +q will move parallel to electric field $\vec{E}$ if it is free to move. The acceleration $\vec{a}$ developed is given as

$$\vec{F}_e = m\vec{a} \quad \text{------- (2)}$$




Comparing eq(1) and eq(2)

$$m\vec{a} = q\vec{E}$$

$$\vec{a} = \frac{q\vec{E}}{m}$$

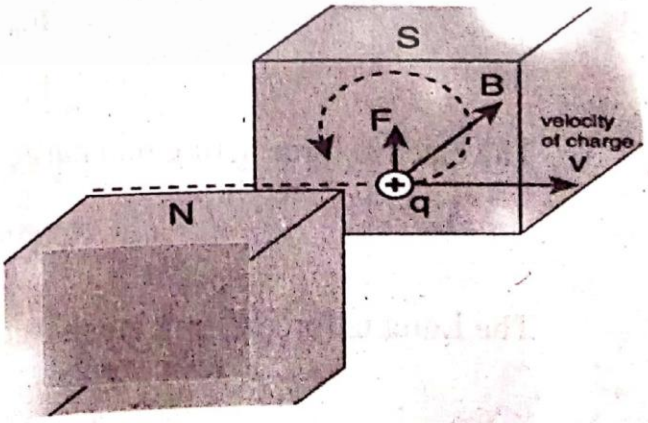
The position of charge at any time can be calculated by using three equations of motion. The electric force always do work on charge when placed inside electric field.

### MOTION OF CHARGED PARTICLE IN MAGNETIC FIELD

The magnetic force $\vec{F}_m$ acts on the charge when a charge + q enters into magnetic field $\vec{B}$ with velocity $\vec{V}$ is given as

$$\vec{F}_m = +q(\vec{V} \times \vec{B})$$

1. The maximum magnetic force acts on +q charge when it enters into magnetic field perpendicularly.
2. The charge moves in circular path under the action of this magnetic force.



❸ The magnetic force which is always perpendicular to the plane defined by $\vec{V}$ and $\vec{B}$ is only deflecting force.

❹ It does not perform any work.

$$W = \vec{F}_m \cdot \vec{S} = \vec{F}_m \cdot \vec{V}\, t$$
$$W = F\, V\, t \cos 90^0$$
$$W = 0$$

**LORENTZ FORCE**

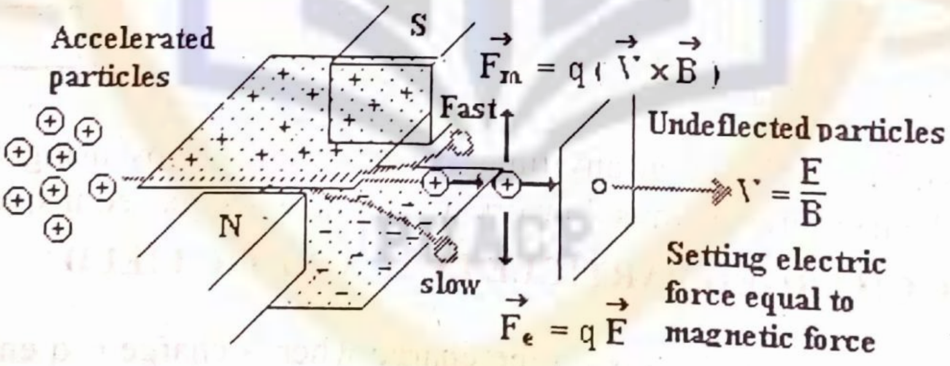
The total force acting on a charge is called Lorentz force when it enters in a region where electric field and magnetic field are applied at the same time. The value of Lorentz force is given as

$$\vec{F} = \vec{F}_e + \vec{F}_m$$
$$\vec{F} = q\vec{E} + q(\vec{V} \times \vec{B})$$

## Q.5 Discuss application of Lorentz force in velocity selector?

**LORENTZ FORCE AND VELOCITY SELECTOR**

The particle accelerator is an application of Lorentz force. Consider a positively charged particle having charge +q. This particle is moving with velocity $\vec{V}$ enters in a region where electric field $\vec{E}$ and magnetic field $\vec{B}$ are applied perpendicularly to each other as shown in diagram.




The magnetic force acting on charge q is given as

$$\vec{F}_m = q(\vec{V} \times \vec{B}) = q\, V\, B \sin 90^0\, \hat{n}$$
$$F_m = q\, V\, B$$

The electric force acting on charge q is given as

$$\vec{F}_e = q\vec{E}$$

The Lorentz force acting on charge q is sum of electric force $\vec{F}_e$ and magnetic force

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$$\vec{F} = \vec{F}_e + \vec{F}_m$$

$$\vec{F} = q\vec{E} + q(\vec{V} \times \vec{B})$$

The electric force $\vec{F}_e$ and magnetic force $\vec{F}_m$ acting on charge q are in opposite direction to each other. The Lorentz force will become zero when electric field and magnetic field are so adjusted that their magnitudes becomes equal. The crossed $\vec{E}$ and $\vec{B}$ fields acts as velocity selector.

$$F_e = F_m$$

$$qE = qVB$$

$$V = \frac{E}{B}$$

Only particles with speed V = E/B pass through the region un deflected by the two fields while particles with other velocities are deflected.

This velocity is independent of charge or mass of particles. The particles with a chosen speed can be isolated from beam by using velocity selector method.

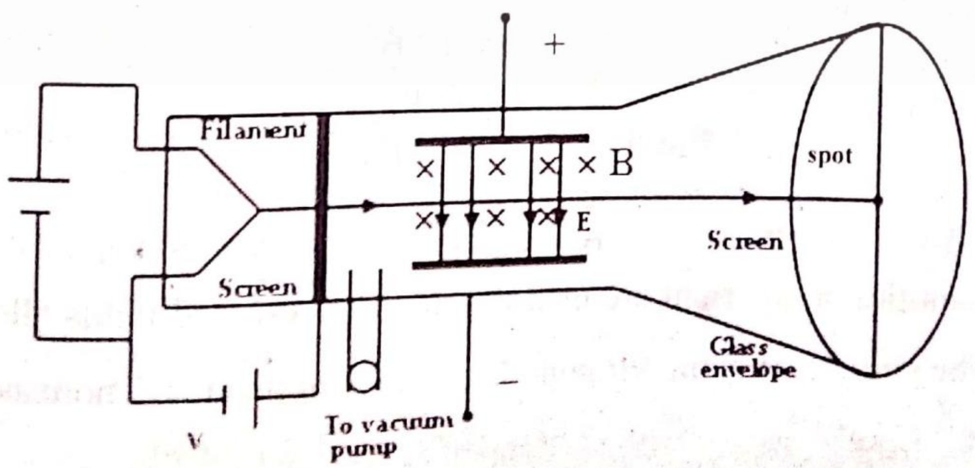
## Q.6 Discuss application of Lorentz force in measurement of charge to mass ratio of electron?

### LORENTZ FORCE & CHARGE TO MASS RATIO OF ELECTRON

The charge to mass ratio is determined by applying Lorentz force. The electric field $\vec{E}$ and magnetic field $\vec{B}$ produce a force on a charged particle. These fields are called crossed fields when they are perpendicular to each other.

Consider a simplified version of apparatus of Thomson's experiment in which an electron moves through crossed fields.

The charged particles such as electrons are emitted by a hot filament at the rear of the evacuated tube and are accelerated by an applied potential difference V as shown in diagram.



These electrons form a narrow beam after passing through a slit. After that they pass through a region of crossed fields $\vec{E}$ and $\vec{B}$. Finally, these electrons are headed towards a fluorescent screen, where they produce a spot of light .The forces on the charged particles in the crossed-fields region can deflect them from the center of the screen.

By controlling the magnitudes and directions of the fields, Thomson could thus control where the spot of light appeared on the screen.

The force acting on a negatively charged particle due to an electric field is directed opposite the field. Thus electrons are forced up the page by electric field as shown in fig and down the page by magnetic field $\vec{B}$. It means forces are in opposition.

Thomson's procedure was equivalent to the following series of steps

❶ Set E = 0 and B = 0 and note the position of the spot on the screen due to the undeflected beam

❷ Turn on $\vec{E}$ and measure the resulting beam deflection.

❸ Maintaining $\vec{E}$ now turn on $\vec{B}$ and adjust its value until the beam returns to the undeflected position.

The deflection of the particle at the far end of the plates is given as

$$y = \frac{e\,EL^2}{2mV^2} \qquad \text{----------(1)}$$

Where V is speed of particle, m is its mass, e is its charge and L is length of the plates. The two deflecting forces cancel each other when electric field and magnetic field are adjusted.

$$F_e = F_m$$

$$q\,E = q\,V\,B\,\text{Sin}90^0$$

$$q\,E = q\,V\,B$$

$$V = \frac{E}{B} \qquad \text{----------(2)}$$

Thus crossed fields allow us to measure the speed of the charged particles passing through them.

Put eq(2) in eq(1)

$$y = \frac{e\,E\,L^2\,B^2}{2\,m\,E^2}$$

$$2\,m\,E^2\,y = e\,E\,L^2\,B^2$$

$$\frac{e}{m} = \frac{2\,y\,E}{L^2\,B^2}$$

All quantities on the right are measurable. Thus crossed fields allow us to measure ratio $\frac{e}{m}$ of the particles moving through Thomson's apparatus. Thomson's value for e/m was $1.7 \times 10^{11}$ C/kg.

**Q.5. Derive an expression for magnetic torque acting on a rectangular loop of wire carrying current placed in applied magnetic field?**

## MAGNETIC TORQUE

The magnetic field is generated around the wire in the form of circles when current flows through a loop of wire. The magnetic force develops when such a current carrying loop is placed in applied magnetic field due to interaction between applied magnetic field and generated magnetic field.

This magnetic force tends to rotate the loop about an axis of rotation. This turning effect produced in a coil is called magnetic torque and is the basic principle of rotation of electric motor, galvanometer, voltmeter etc.

## MAGNETIC TORQUE ON CURRENT CARRYING LOOP PLACED IN APPLIED MAGNETIC FIELD

Consider a rectangular loop ABCDA having length "b" and breadth "a" through which current I flows in clockwise direction as shown in fig.

1. The loop is placed in uniform applied magnetic field B directed along y–axis.
2. The area of loop is A = ab.
3. The loop is free to rotate about an axis of rotation which is parallel to along z-axis.

## DIRECTION OF PLANE OF LOOP

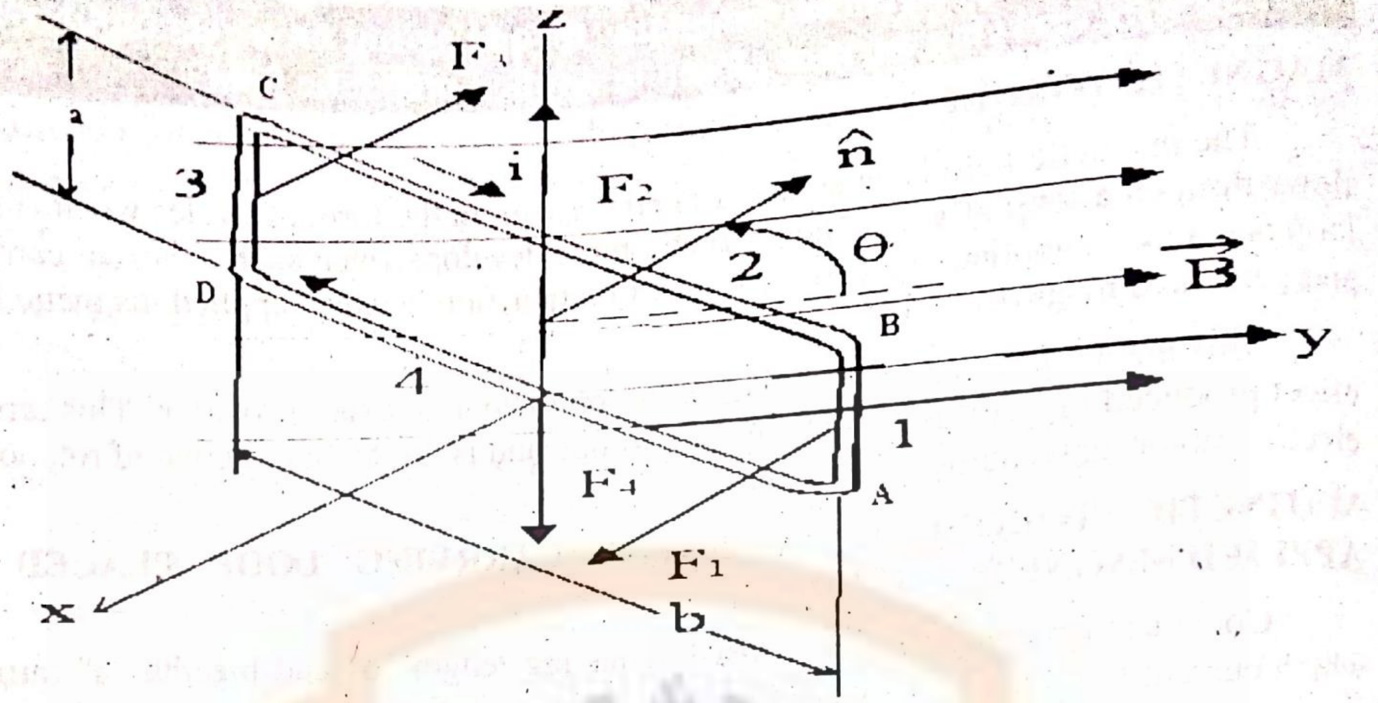
The direction of plane of loop is represented by a unit vector $\hat{\mathbf{n}}$ that is normal to the plane and makes angle θ with magnetic field $\vec{\mathbf{B}}$. The direction of unit vector $\hat{\mathbf{n}}$ is given by right hand rule stated as

The thumb indicates the direction of unit vector if fingers of your right hand are along the direction of the current in the side of a loop.

## MAGNETIC FORCE ACTING ON EACH SIDE

The current carrying loop has four straight sides.

1. The side AB is perpendicular to magnetic field $\vec{\mathbf{B}}$ and directed along z–axis.
2. The side BC makes angle $(90^0 - \theta)$ with magnetic field $\vec{\mathbf{B}}$.
3. The sides CD is perpendicular to magnetic field $\vec{\mathbf{B}}$ and directed along z –axis.
4. The side DA makes angle $(90^0 + \theta)$ with magnetic field $\vec{\mathbf{B}}$.



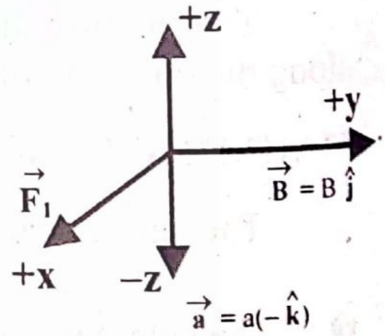

The magnetic force acts on each side of the loop due to interaction between applied magnetic field and generated magnetic field.

The value of magnetic force acting on each side of loop can be calculated by using formula

$$\vec{F} = I(\vec{L} \times \vec{B})$$

**MAGNETIC FORCE ON SIDE AB**

For side AB in which current flows along negative z-axis




$$\vec{L} = \vec{a} = a(-\hat{k})$$

$$\vec{B} = B\hat{j}$$

$$\vec{F}_1 = ?$$

The generated magnetic force in side AB is

$$\vec{F}_1 = -IaB(\hat{k} \times \hat{j})$$

$$\vec{F}_1 = IaB\hat{i}$$

The force $\vec{F}_1$ acts parallel to +x-axis. The magnitude of force is

$$F_1 = IaB \sin 90^\circ = IaB$$

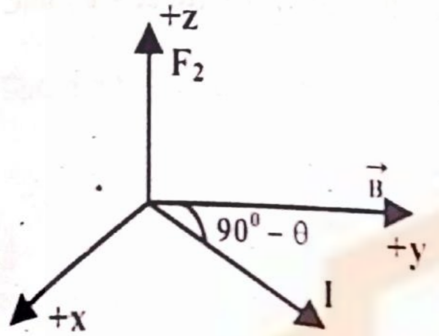
## MAGNETIC FORCE ON SIDE BC

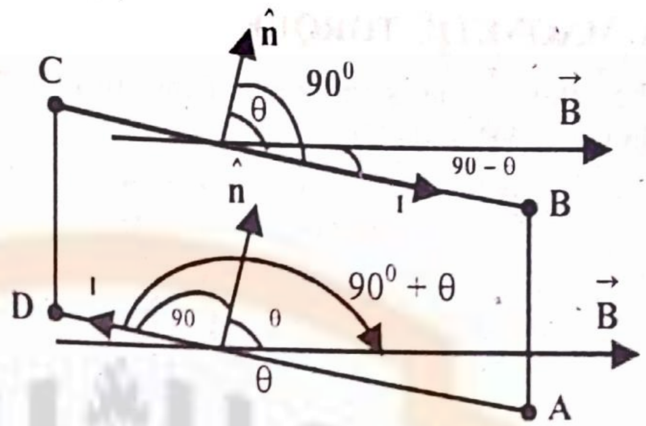
The magnitude of generated magnetic force for side BC is

$$F_2 = I\,b\,B\,\text{Sin}(90^\circ - \theta)$$

$$F_2 = I\,b\,B\,\text{Cos}\theta$$

The direction of force $F_2$ is along +z-axis according to right hand rule



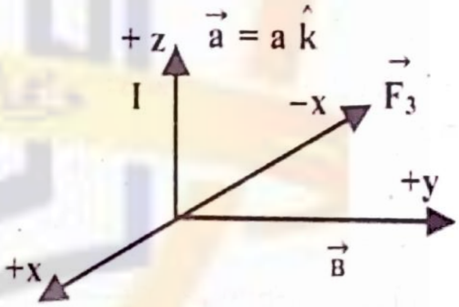

## MAGNETIC FORCE ON SIDE CD

For side CD

$$\vec{L} = \vec{a} = a\,\hat{k}$$

$$\vec{B} = B\,\hat{j}$$

$$\vec{F}_3 = ?$$




The generated magnetic force in side CD is

$$\vec{F}_3 = I\,a\,B(\hat{k} \times \hat{j})$$

$$\vec{F}_3 = -I\,a\,B\,\hat{i}$$

The force $\vec{F}_3$ acts parallel to − x-axis. The magnitude of force is

$$F_3 = IaB\,\text{Sin}90^\circ = IaB$$

## MAGNETIC FORCE ON SIDE DA

The magnitude of generated magnetic force in side DA is

$$F_4 = I\,b\,B\,\text{Sin}(90^\circ + \theta)$$

$$F_4 = I\,b\,B\,\text{Cos}\theta$$

The direction of force $F_4$ is along −z-axis according to right hand rule. When current is reversed in a wire, the direction of force also reverses.

The forces $F_2$ and $F_4$ are equal in magnitude (IbB Cosθ) but opposite in direction. These both forces are directed along z-axis. The net force is zero because their line of action is same. It means torque is zero due to forces $F_2$ and $F_4$.

The forces $F_1$ and $F_3$ are also equal in magnitude (IaB) but opposite in direction but their line of action is not same. Hence these forces $F_1$ and $F_2$ form a couple and give rise to magnetic torque. The magnetic torque tends to rotate the loop about an axis of rotation in clockwise direction.

## TOTAL MAGNETIC TORQUE

The total torque is the sum of the torques produced due to both forces $F_1$ and $F_3$ acting on sides AB and CD.

$$\vec{\tau}_m = \vec{\tau}_1 + \vec{\tau}_2$$

$$\vec{\tau}_m = \vec{r} \times \vec{F}_1 + \vec{r} \times \vec{F}_3$$

$$\vec{\tau}_m = r F_1 \sin\theta + r F_3 \sin\theta$$

$$\vec{\tau}_m = r (I\, a\, B) \sin\theta + r (I\, a\, B) \sin\theta$$

$$\vec{\tau}_m = 2 r I\, a\, B \sin\theta$$

Put r = b/2 because z-axis divide the length of loop into half and passes through center.

$$\vec{\tau}_m = 2\left(\frac{b}{2}\right) I\, a\, B \sin\theta$$

The magnitude of torque is

$$\tau_m = I\, a\, b\, B \sin\theta$$

## TOTAL MAGNETIC TORQUE FOR N TURNS

The total magnetic torque acting on loop having N turns is

$$\tau_m = N I (a\, b) B \sin\theta$$

$$\tau_m = N I A B \sin\theta \qquad [A = ab = \text{area of loop}]$$

This torque is called magnetic torque.

$$\tau_m = N I A B \sin\theta$$

$$\tau_m = \mu B \sin\theta$$

Where $\mu = NIA$ is called magnetic dipole moment.

This value of torque holds for all kinds of loops either rectangular or circular.

**Q.6. What is magnetic dipole moment? Calculate work done and magnetic potential energy needed for the orientation of magnetic dipole moment in magnetic field?**

## MAGNETIC DIPOLE MOMENT

The product of current and area of a current carrying loop is called magnetic dipole moment. It is denoted by μ. Its SI unit is J/T

Mathematically it is written as

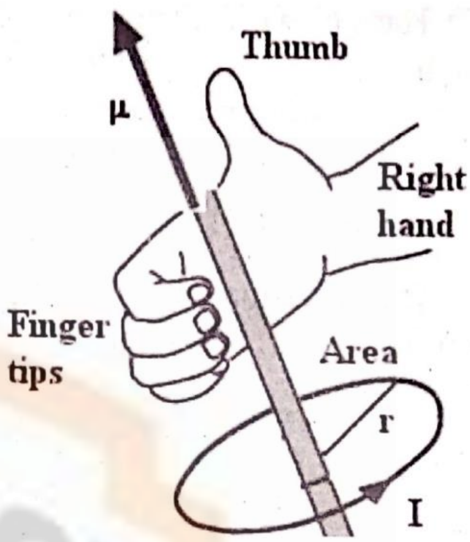
$$\mu = I A$$

The magnetic dipole moment if loop has N turns

$$\mu = N I A$$

Many physical systems have magnetic dipole moments like earth, bar magnets, current loops, atoms, nuclei and elementary particles.

The magnetic dipole moment is a vector quantity whose direction is given by right hand rule.




## RIGHT HAND RULE

Hold the current carrying loop in right hand in such a way that fingertips are along the direction of the current, then erect thumb indicates the direction of magnetic moment as shown in fig. The direction of magnetic dipole moment and normal to loop is same.

## RELATION BETWEEN MAGNETIC TORQUE & MAGNETIC DIPOLE MOMENT

The magnetic torque acts on the loop when current carrying loop is placed in applied magnetic field and tends to rotate the loop.

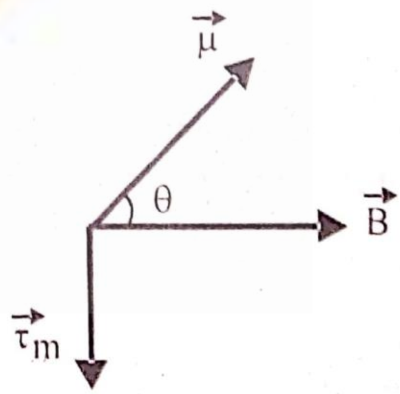
The magnitude of this magnetic torque is given as

$$\tau_m = \mu B \sin\theta$$

In vector form torque is written as

$$\vec{\tau}_m = \vec{\mu} \times \vec{B}$$

The direction of magnetic torque is given by right hand rule stated as




❶ Place close-fisted right hand at a location where tails of magnetic dipole moment $\vec{\mu}$ and magnetic field vector $\vec{B}$ are joined together in such a way that fingertips are along the direction of rotation through smaller angle,

❷ The stretched thumb indicates the direction of magnetic torque.

## WORK DONE

The amount of work done by magnetic field in turning the magnetic dipole moment through angle $d\theta$ in rotational motion is

$$dW = \vec{\tau}_m \cdot \vec{d\theta}$$

The net work done in turning the magnetic dipole from initial angle $\theta_0$ to final angle $\theta$

$$W = \int_{\theta_0}^{\theta} dW = \int_{\theta_0}^{\theta} \vec{\tau}_m \cdot \vec{d\theta}$$

The direction of magnetic torque $\vec{\tau}_m$ and angular displacement $\vec{d\theta}$ is opposite according to right hand rule.

$$W = \int_{\theta_0}^{\theta} \tau_m \, d\theta \operatorname{Cos}180^\circ$$

$$W = -\int_{\theta_0}^{\theta} \tau_m \, d\theta = -\int_{\theta_0}^{\theta} \mu B \operatorname{Sin}\theta \, d\theta$$

$$W = -\mu B \left| -\operatorname{Cos}\theta \right|_{\theta_0}^{\theta}$$

$$W = \mu B (\operatorname{Cos}\theta - \operatorname{Cos}\theta_o)$$

## MAGNETIC POTENTIAL ENERGY

The negative work done is equal to change in magnetic potential energy

$$\Delta U = -W$$

$$\Delta U = -\mu B (\operatorname{Cos}\theta - \operatorname{Cos}\theta_o)$$

$$U(\theta) - U(\theta_0) = -\mu B (\operatorname{Cos}\theta - \operatorname{Cos}\theta_0)$$

Take arbitrary value of $\theta_0 = 90^\circ$ and choose magnetic potential energy be zero at that angle. i.e. $U(90^\circ) = 0$

Then at any angle magnetic potential energy is

$$U(\theta) - 0 = -\mu B (\operatorname{Cos}\theta - \operatorname{Cos}90^\circ)$$

$$U(\theta) = -\mu B \operatorname{Cos}\theta$$

$$U(\theta) = -\vec{\mu} \cdot \vec{B}$$

The magnetic potential energy will be minimum when magnetic dipole moment $\mu$ and magnetic field B are parallel i.e. $\theta = 0$

$$U = -\mu B \operatorname{Cos}\theta = -\mu B \operatorname{Cos}0$$

$$U = -\mu B$$

The magnetic potential energy will be maximum when magnetic dipole moment $\mu$ and magnetic field B are anti parallel i.e. $\theta = 180^o$

$$U = -\mu B \cos 180^o$$

$$U = +\mu B$$

# SHORT QUESTIONS

**Q:1 Can electromagnetic wave be deflected by magnetic field.**

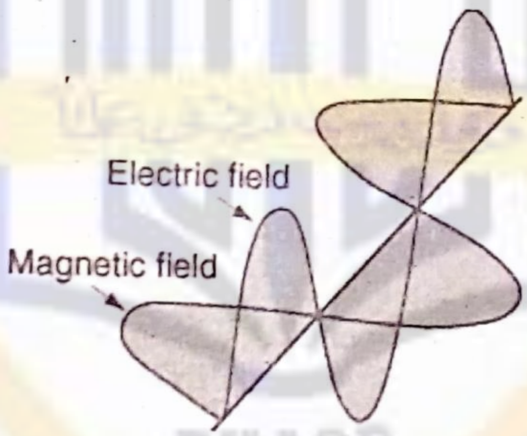
**Ans:** The electromagnetic waves are charge-less photons so cannot be deflected by magnetic field.

**Q:2 Can a charged particle at rest be set in motion by the action of a magnetic field.**

**Ans:** The charged particle at rest cannot be set in motion by the action of a magnetic field because magnetic force $F_m = q V B$ only acts on charge having velocity.

**Q:3 Are the electromagnetic waves longitudinal or transverse.**

**Ans:** The electromagnetic waves are combination of electric field vectors and magnetic field vectors which perpendicular to each other. Hence electromagnetic waves are transverse in nature.




**Q:4 What is angle between direction of flow of current and external magnetic field to have magnetic force of maximum value.**

**Ans:** The magnetic force acting on current carrying wire placed in magnetic field is $F_m = I L B \sin\theta$. The angle between direction of flow of current and external magnetic field to have magnetic force of maximum value is $90^0$.

**Q:5 The equation $\vec{\tau} = \vec{\mu} \times \vec{B}$ shows that there is no torque on a current loop in an external magnetic field if the angle between the axis of the loop and the field is (a) $0^0$ (b) $180^0$. Discuss the nature of the equilibrium for these two positions.**

**Ans:** Since $\vec{\tau} = \vec{\mu} \times \vec{B} = \mu B \sin\theta \hat{n}$. The value of torque is zero when $\theta = 0^0$ or $\theta = 180^0$. Hence equilibrium is stable. There will be no rotation.

**Q:5 Does $\vec{F} = I(\vec{L} \times \vec{B})$ hold for a straight wire whose cross-section varies irregular along its path.**

**Ans:** Yes, the relation $\vec{F} = I(\vec{L} \times \vec{B})$ holds for a straight wire whose cross-section varies irregular along its path because strength of current is independent of cross-section.

**Q:6 Is magnetic force conservative or non-conservative. Could we define a magnetic potential energy as we defined electric or gravitational potential energy?**

**Ans:** The magnetic force is conservative. Yes, we can define a magnetic potential energy as we defined electric or gravitational potential energy.

**Q:7 A beam of electrons can be deflected either by electric field or by a magnetic field. Is one method better than other?**

**Ans:** The deflection of electrons beam by electric field is better than deflection by magnetic field because electric field is stronger than magnetic field and deflection by magnetic field is conditional.

**Q:8 If a moving electron is deflected sideways in passing through a certain region of space, can we be sure that a magnetic field exists in that region.**

**Ans:** The moving electron is deflected either by magnetic field or electric field.

When a moving electron is deflected sideways in passing through a certain region of space, we are sure that magnetic field exists in that region.

**Q:9 If an electron is not deflected in passing through a certain region of space, can we be sure that there is no magnetic field in that region.**

**Ans:** The electron in certain region is deflected by magnetic force. The magnetic force will be zero if an electron moves parallel or anti-parallel in a region of magnetic field. It means an electron will not be deflected even that field is present.

Therefore, when an electron is not deflected in passing through a certain region of space, we cannot be sure that there is a magnetic field in that region.

**Q:10 How could you rule out that the forces between two magnets are electrostatic forces.**

**Ans:** The origin of magnetism is due to motion of electrons in orbits of atoms. These are moving charges. Hence forces between two magnets are not electrostatics.

**Q:11 Imagine you are sitting in a room with your back to one wall and that an electron beam traveling horizontally from back wall to the front wall is deflected to your right. What is the direction of the uniform magnetic field that exists in the room?**

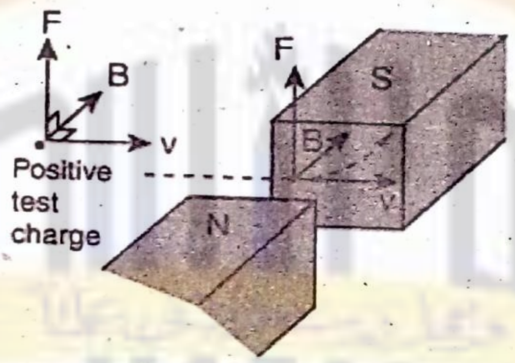
**Ans:** According to right hand rule, the magnetic field is directed from top of room towards floor of room.

**Q:12 Why do we not simply define the direction of the magnetic field $\vec{B}$ to be the direction of the magnetic force that acts on a moving charge.**

**Ans:** The magnetic force acting on charge moving in applied magnetic field is given as $\vec{F} = q(\vec{V} \times \vec{B})$. The magnetic force is always perpendicular to the plane defined by $\vec{V}$ and $\vec{B}$. Therefore, we cannot define the direction of the magnetic field $\vec{B}$ to be the direction of the magnetic force that is acting on a moving charge.

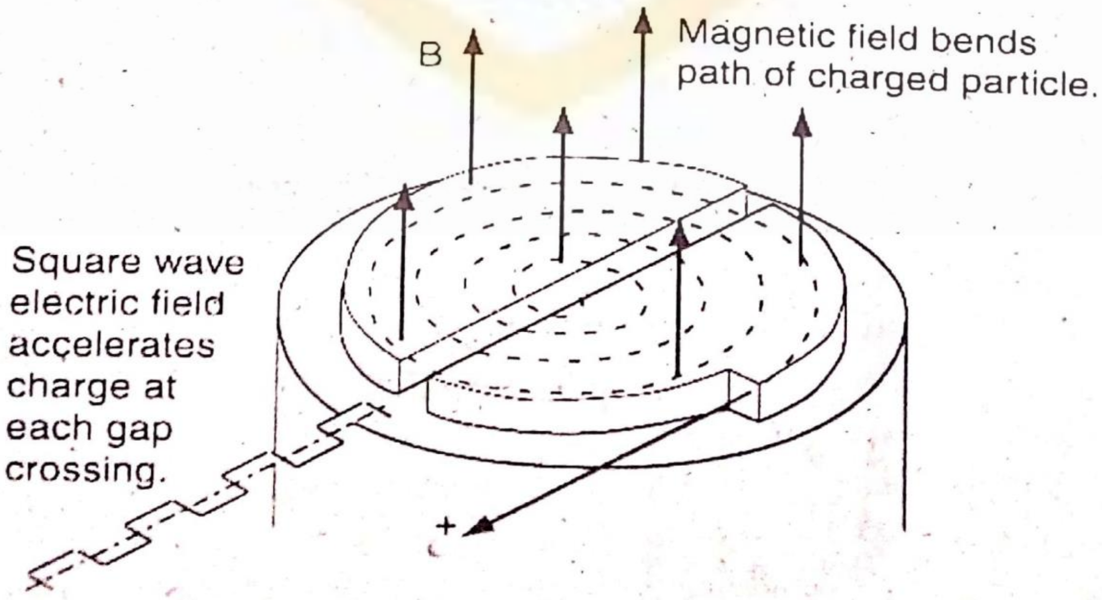
**Q:13 Of the three vectors in the equation $\vec{F} = q(\vec{V} \times \vec{B})$ which pairs are always right angle which may have any angle.**

**Ans:** The magnetic force is always perpendicular to plane defined by $\vec{V}$ and $\vec{B}$. Therefore, the pair $(\vec{F},\vec{V})$ and pair $(\vec{F},\vec{B})$ are always perpendicular and pair $(\vec{V},\vec{B})$ may have any angle.




**Q:14 What is the function of electric field and magnetic field in cyclotron?**

**Ans:** The magnetic field is used to bend the path of ions in semicircle. The electric field is used to accelerate ions between dees. The ion is accelerated again and again in gaps between dees and gets energy in steps.



# SAMPLE PROBLEMS

**P.1** **A strip of copper 150 μm thick is placed in a magnetic field 0.65 T perpendicular to the plane of strip, and a current 23 A is set up in the strip. What Hall pd would appear across the width of strip of there were one charge carries per atom?**

**Sol.**

**DATA**

For copper strip

$n = 8.49 \times 10^{28}$ electrons/ms

$B = 0.65$ T

$t = 150\ \mu m = 150 \times 10^{-6}$ m

$I = 23$ A

**CALCULATIONS**

$$V = \frac{IB}{net}$$

$$V = \frac{23 \times 0.65}{8.49 \times 10^{28} \times 1.6 \times 10^{-19} \times 150 \times 10^{-6}} = 7.3 \times 10^{-6}\ \text{m/s}$$

**P.2** **A straight horizontal segment of Cu wire carries a current of 28 A. What is the magnitude and direction of the magnetic field needed to float the wire to balance its weight? Its linear mass density is 46.6 g/m.**

**Sol.**

**DATA**

$I = 28$ A

$\frac{m}{L} = 46.6$ g/m

$B = ?$

**CALCULATIONS**

Since weight force is balanced by magnetic force.

$$F_m = F_g$$

$$ILB = mg$$

$$B = \frac{(m/L)g}{I}$$

$$B = \frac{46.6 \times 10^{-3} \times 9.8}{28} = 0.0163\ \text{T}$$

**P.3** **A rectangular loop of wire consisting of nine turns and having width a = 0.103 m and length b = 0.685 m is attached to one pan of a balance in magnetic field B. A current 0.224 A is set up in wire and is found that to restore balanced condition a mass 13.7 g must be added to other pan. Find B.**

**Sol.**

**DATA**

$N = 9$

$I = 0.224\ A$

$a = 0.103\ m$

$b = 0.685\ m$

$m = 13.7\ g$

$B = ?$

**CALCULATIONS**

$$F_m = F_g$$

$$NIaB = mg$$

$$B = \frac{m\,g}{N\,I\,a}$$

$$B = \frac{13.7 \times 10^{-3} \times 9.8}{9 \times 0.224 \times 0.103} = 0.64\ T$$

**P.4** **(a) What is magnetic dipole moment of the coil which is 2.1 cm high and 1.2 cm wide with 250 turns as used in previous problem assuming that it carries current of 85 μA (b) The magnetic dipole moment of the coil is lined up with external magnetic field of 0.85 T. How much work would be done by external agent to rotate the coil through 180°.**

**Sol.**

**DATA**

$A = 2.1 \times 1.2 = 2.52\ cm^2 = 2.52 \times 10^{-4}\ m^2$

$N = 250$

$I = 85\ \mu A$

$B = 0.85\ T$

**CALCULATIONS**

**(a)** $\mu = ?$

$$\mu = N\,I\,A$$

$$\mu = 250 \times 85 \times 10^{-6} \times 2.52 \times 10^{-4}$$

$$\mu = 5.36 \times 10^{-6}\ J/T$$

**(b)** W = ?

$$W = \mu B(\cos\theta - \cos\theta_o)$$
$$W = \mu B(\cos 180^\circ - \cos\theta_o)$$
$$W = \mu B(\cos 180^\circ - \cos 0^0)$$
$$W = \mu B(-1 - 1)$$
$$W = -2\mu B$$
$$W = -2 \times 5.36 \times 10^{-6} \times 0.85$$
$$W = -9.1\ \mu J$$

# EXERCISE PROBLEMS

**P.1 An electron in a TV camera tube is moving at $7.2 \times 10^6$ m/s in a magnetic field 83 mT (a) Without knowing the direction of the field what could be the greatest and least magnitudes of the force of electron could feel due to field (b) At one point the acceleration of electron is $4.9 \times 10^{+16}$ m/s$^2$. What is an angle b/w electron velocity and magnetic field?**

**Sol.**

**DATA**

$$q = 1.6 \times 10^{-19} C$$
$$V = 7.2 \times 10^6 m/s$$
$$B = 83 mT$$
$$F_m = ?\ (\text{max. value})$$
$$F_m = ?\ (\text{min. value})$$
$$a = 4.9 \times 10^{16} m/s^2$$
$$\theta = ?$$

**CALCULATIONS**

(a) For maximum force $\theta = 90^\circ$

$$F_m = q V B \sin\theta$$
$$(F_m)_{max} = 1.6 \times 10^{-19} \times 7.2 \times 10^6 \times 83 \times 10^{-3} \sin 90^0$$
$$(F_m)_{max} = 9.56 \times 10^{-14} N$$

For minimum force ($\theta = 0^\circ$)

$$F_m = q VB \sin\theta$$
$$(F_m)_{min} = 0$$

**(b)** For a point where acceleration is $4.9 \times 10^{+6}$m/s$^2$, we have

$$F_m = q V B \sin\theta$$
$$m a = q V B \sin\theta$$

$$\theta = \text{Sin}^{-1}\left(\frac{m\,a}{qVB}\right)$$

$$\theta = \text{Sin}^{-1}\left(\frac{9.1\times10^{-31}\times4.9\times10^{16}}{1.6\times10^{-19}\times7.2\times10^{6}\times83\times10^{-3}}\right)$$

$$\theta = 28^{o}$$

**P.2** **An electric field of 1.5 kV/m and a magnetic field of 0.44 T acts on a moving electron to produce no force (a) Calculate the minimum electron speed V. Draw the vectors $\vec{E}$, $\vec{B}$ and $\vec{V}$.**

**Sol.**

**DATA**

$E = 1.5\times10^{3}\,V/m$

$B = 0.44\,T$

$q = 1.6\times10^{-19}C$

$m = 9.1\times10^{-31}\,kg$

$V = ?$

**CALCULATIONS**

Since no force acts on electron, the electric force and magnetic force is equal or balance each other.

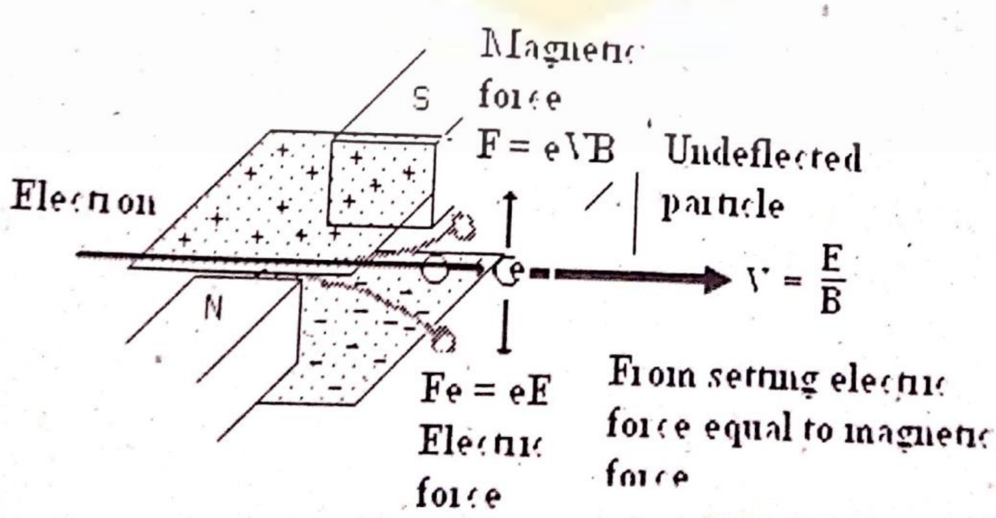
$$F_c = F_m$$

$$q\,E = q\,V\,B$$

$$V = \frac{E}{B}$$

$$V = \frac{1.5\times10^{3}}{0.44} = 3409\ m/s$$

**PLOT OF VECTORS $\vec{E}$, $\vec{B}$ AND $\vec{V}$**



**P.3** **A proton travelling at 23° with respect to a magnetic field of 2.63 mT experiences a magnetic force of $6.48 \times 10^{-17}$ N. Calculate the speed and KE of proton in eV.**

**Sol.**

**DATA**

$$q = 1.6 \times 10^{-19} C$$
$$B = 2.63 \text{ mT}$$
$$\theta = 23^\circ$$
$$F_m = 6.48 \times 10^{-17} N$$
$$V = ?$$
$$KE = ?$$

**CALCULATIONS**

$$F_m = q V B \sin\theta$$
$$V = \frac{F_m}{q B \sin\theta}$$
$$V = \frac{6.48 \times 10^{-17}}{1.6 \times 10^{-19} \times 2.63 \times 10^{-3} \sin 23^0}$$
$$V = 394113 \text{ m/s}$$

KE of proton

$$K.E. = \frac{1}{2} mV^2$$
$$K.E. = \frac{1}{2} \times 1.67 \times 10^{-27} (394113)^2$$
$$KE = 1.3 \times 10^{-16} J$$
$$KE = \frac{1.3 \times 10^{-16}}{1.6 \times 10^{-19}} = 810 \text{ eV}$$

**P.4** **A cosmic ray proton impinges on the earth near the equator with a vertical velocity of $2.8 \times 10^7$m/s. Assume that the horizontal component of earth magnetic field at equator is 30 μT. Calculate the ratio of magnetic force on proton to the gravitational force on it?**

**Sol.**

**DATA**

$$m = 1.67 \times 10^{-27} kg$$
$$V = 2.8 \times 10^7 m/s$$
$$B = 30 \times 10^{-6} T$$
$$q = 1.6 \times 10^{-19} C$$
$$\theta = 90^\circ$$
$$\frac{F_m}{F_g} = ?$$

**CALCULATIONS**

$$\frac{F_m}{F_g} = \frac{q\,V\,B\,\text{Sin}\theta}{mg}$$

$$\frac{F_m}{F_g} = \frac{1.6\times10^{-19}\times2.8\times10^{7}\times30\times10^{-6}\times\text{Sin}90^{0}}{1.67\times10^{-27}\times9.8}$$

$$\frac{F_m}{F_g} = 8.2\times10^{9}$$

**P.5** **An electron in uniform magnetic field has a velocity $\vec{V} = (40\,\hat{i} + 35\,\hat{j})$ km/s experiences a force $\vec{F} = (-4.2\,\hat{i} + 4.8\,\hat{j})$ fN. If $B_x = 0$, calculate the magnetic field.**

**Sol.**

**DATA**

$q = 1.6\times10^{-19}$C

$\vec{V} = (40\,\hat{i} + 35\,\hat{j})$ km/s

$\vec{F} = (-4.2\,\hat{i} + 4.8\,\hat{j})$ fN

$B_x = 0$

$\vec{B} = ?$

**CALCULATIONS**

$$\vec{F}_m = q(\vec{V}\times\vec{B})$$

$$(-4.2\,\hat{i}+4.8\,\hat{j})\times10^{-15} = 1.6\times10^{-19}\left[(40\,\hat{i}+35\,\hat{j})\times10^{3}\times(B_x\,\hat{i}+B_y\,\hat{j}+B_z\,\hat{k})\right]$$

$$(-4.2\,\hat{i}+4.8\,\hat{j})\times10^{-15} = 1.6\times10^{-16}\left[(35B_z\,\hat{i}-40\,B_z\,\hat{j}+40\,B_y\,\hat{k})\right]$$

Comparing coefficient of unit vectors

$B_Z = -0.75$

$B_y = 0$

The net magnetic field is

$$\vec{B} = B_x\,\hat{i}+B_y\,\hat{j}+B_z\,\hat{k}$$

$$\vec{B} = 0\,\hat{i}+0\,\hat{j}-0.75\,\hat{k}$$

**P.6** **(a) In a magnetic field with B = 0.5 T for what path radius will an electron circulate at 0.10 the speed of light (b) What will be its KE in eV. Ignore small relativistic effects.**

**Sol.**

**DATA**

$$q = 1.6 \times 10^{-9}\,C$$
$$B = 0.5\,T$$
$$m = 9.1 \times 10^{-31}\,kg$$
$$V = 0.1c$$
$$r = ?$$

**CALCULATIONS**

**(a)**

$$F_c = F_m$$
$$\frac{mV^2}{r} = q\,V\,B$$
$$r = \frac{mV}{q\,B}$$
$$r = \frac{9.1 \times 10^{-31} \times 0.1 \times 3 \times 10^{8}}{1.6 \times 10^{-19} \times 0.5}$$
$$r = 3.4 \times 10^{-4}\,m$$

**(b)**

$$KE = \frac{1}{2}mV^2$$
$$KE = \frac{1}{2}9.1 \times 10^{-31}(0.1 \times 3 \times 10^{8})^2$$
$$KE = 4.09 \times 10^{-16}\,J$$
$$KE = \frac{4.09 \times 10^{-16}}{1.6 \times 10^{-19}}\,eV = 2.6\,keV$$

**P.7** **Show that in terms of Hall electric field E and current density J, the number of charge carries per unit volume is given by $n = \frac{J\,B}{eE}$.**

**Sol.**

**DATA**

$$n = \frac{I\,B}{eV\,t}$$
$$n = \frac{J\,A\,B}{eV\,t} \qquad [J = I/A]$$
$$n = \frac{J.(L\,t)\,B}{eV\,t} = \frac{J\,B}{eV/L} = \frac{J\,B}{eE}$$

**P.8** **A 1.15 kg copper rod rests on two horizontal rails 95 m apart and carries a current of 53.2 A from one rail to other. The coefficient of static frication is 0.58. Find the smallest magnetic field that would cause the bar to slide.**

**Sol.**

**DATA**

$m = 1,15 \text{ kg}$

$L = 95 \text{ m}$

$\mu_s = 0.58$

$I = 53.2 \text{A}$

$B = ?$

**CALCULATIONS**

$$F_m = F$$

$$I L B = \mu_s mg$$

$$B = \frac{\mu_s mg}{I L}$$

$$B = \frac{0.58 \times 1.15 \times 9.8}{53.2 \times 95} = 1.3 \times 10^{-3} \text{ T}$$

**P.9** **A circular coil of 160 turns has a radius of 1.93 cm. (a) Calculate the current that results in a magnetic moment of 2.3 A–m$^2$ (b) Find the max. torque on the coil that can be experienced in field of 34.6 mT.**

**Sol.**

**DATA**

$N = 160$

$r = 1.93 \text{ cm}$

$\mu = 2.3 \text{ A–m}^2$

$B = 34.6 \text{ mT}$

$I = ?$

$\tau = ?$ (max value for which $\theta = 90°$)

**CALCULATIONS**

**(a)**

$$\mu = N I A = N I \pi r^2$$

$$I = \frac{\mu}{N \pi r^2}$$

$$I = \frac{2.3}{160 \times 3.14 (1.93 \times 10^{-2})^2} = 12.4 \text{ A}$$

**(b)**

$$\tau = N I A B \sin\theta$$

$$\tau = 160(12.4)\pi(1.93 \times 10^{-2})^2 (34.6) \sin 90°$$

$$\tau = 80 \text{ N–m}$$

**P.10** **The magnetic dipole moment of earth is $8.0 \times 10^{22}$ J/T. Assume that this is produced by charges flowing in the molten outer core of earth. If the radius of circular path is 3500 km. Calculate current.**

**Sol.**

**DATA**

$N = ?$

$\mu = 8.0 \times 10^{22}$ J/T

$r = 3500$ km

$I = ?$

**CALCULATIONS**

$$\mu = N I A$$

$$\mu = N I \pi r^2$$

$$I = \frac{\mu}{N \pi r^2}$$

$$I = \frac{8.0 \times 10^{22}}{1 \times 3.14 (3500 \times 10^3)^2} = 2.08 \times 10^9 \text{ A}$$

**P.11** **A circular wire loop whose radius 16 cm carries a current of 2.58 A. It is placed so that the normal to its plane makes angle 41° with field 1.2 T. (a) Calculate the magnetic dipole moment of the loop. (b) Find the torque on the loop.**

**Sol.**

**DATA**

$\theta = 41^\circ$

$r = 16 \text{ cm} = 16 \times 10^{-2}$ m

$I = 2.58$ A

$B = 1.2$ T

**CALCULATIONS**

(a) $$\mu = N I A$$

$$\mu = N I \pi r^2$$

$$\mu = 1 \times 2.58(16 \times 10^{-2})^2 \times 3.14 = 0.207 \text{ J/T}$$

(b) $$\tau = N I A B \sin\theta$$

$$\tau = 1 \times 2.58(16 \times 10^{-2})^2 \pi \times 1.2 \times \sin 41^\circ$$

$$\tau = 0.162 \text{ N-m}$$

**P.12** **A circular wire loop having radius 8 cm carries a current of 0.2 A. A unit vector parallel to the dipole moment μ of the loop is given by $0.6\,\hat{i} - 0.8\,\hat{j}$. If the loop is located in $\vec{B} = (0.25\,\hat{i} + 0.3\,\hat{k})$T (a) Find the torque on the loop (b) Find magnetic potential energy of the loop.**

**Sol.**

**DATA**

$$\vec{B} = (0.25\,\hat{i} + 0.3\,\hat{k})\text{T}$$
$$\hat{\mu} = 0.6\,\hat{i} - 0.8\,\hat{j}$$
$$I = 0.2\text{ A}$$
$$r = 8\text{ cm}$$
$$\vec{\tau} = ?$$
$$U_m = ?$$

**CALCULATIONS**

(a)
$$\vec{\mu} = \mu\,\hat{\mu}$$
$$\vec{\mu} = N I A\,\hat{\mu}$$
$$\vec{\mu} = 1 \times 0.2 \times 3.14 \times (8 \times 10^{-2})^2\,\hat{\mu}$$
$$\vec{\mu} = 4.01 \times 10^{-3}\,(0.6\,\hat{i} - 0.8\,\hat{j})\text{ J/T}$$
$$\vec{\mu} = (2.41 \times 10^{-3}\,\hat{i} - 3.20 \times 10^{-3}\,\hat{j})\text{ J/T}$$

The torque acting on the loop

$$\vec{\tau}_m = \vec{\mu} \times \vec{B}$$
$$\vec{\tau}_m = [(2.41\,\hat{i} - 3.20\,\hat{j} + 0\,\hat{k}) \times 10^{-3}] \times [(0.25\,\hat{i} + 0\,\hat{j} + 0.3\,\hat{k})]$$
$$\vec{\tau}_m = 9.1 \times 10^{-4}\,\hat{i} - 7.23 \times 10^{-4}\,\hat{j} \times 8.0 \times 10^{-4}\,\hat{k}$$

(b)
$$U_m = -\vec{\mu}\,.\,\vec{B}.$$
$$U_m = -[(2.41\,\hat{i} - 3.2\,\hat{j} + 0\,\hat{k}) \times 10^{-3}]\,.\,[(0.25\,\hat{i} + 0\,\hat{j} + 0.3\,\hat{k})]$$
$$U_m = 2.41 \times 10^{-3} \times 0.25$$
$$U_m = 6.025 \times 10^{4}\text{N–m}$$

**P.13 At a place earth's magnetic field is 39.0 μT and it is horizontal due north. Calculate magnetic force on a conductor of length 1 m carrying current of 10 mA due east.**

**Sol.**

**DATA**

$B = 39.0\,\mu T = 39 \times 10^{-6}\,T$

$L = 1\,m$

$I = 10\,mA = 10 \times 10^{-3}\,A$

$\theta = 90^0$

$F_m = ?$

**CALCULATIONS**

$$F_m = I L B \sin\theta$$

$$F_m = 10 \times 10^{-3} \times 1 \times 39 \times 10^{-6} \sin 90^0$$

$$F_m = 3.9 \times 10^{-7}\,N$$